



Project title: VR Industrial Training Prototype

Team Name: CaffeineLife

Group Members	
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Submission date: 29 August 2026

1. Project Scope

1.1 Overview

This project builds on a deliberately broken factory VR training prototype provided as the base environment. The work is divided into two parts. “Part A” repairs seven pre-existing defects across the onboarding UI, console interaction logic, baked lighting, and visual or interaction feedback, restoring the environment to a polished, realistic state. “Part B” extends the repaired environment with one original interactive feature which was the safety equipment check station.

1.2 Repair Scope (Part A)

Seven issues were diagnosed and fixed in the provided scene, spanning UI or canvas configuration, toggle logic, lightmap resolution and baking, light probe placement, and interactor gating during onboarding. Each fix is summarized in the table below and illustrated with before and after evidence in Section 2.

No.	Area	Issue	Fix Applied
1.	Onboarding UI	Fade-in screen appeared mid-scene instead of covering the full field of view.	Canvas_Fade, inspector set the Plane Distance to ‘0.1’ and temporarily set the “Canvas Group (Alpha)” to 1 so that you can see the outline of the canva.
2.	Console Labels Toggle	Toggling “Console Labels” deactivated the building instead of only its text labels.	Canvas_Main → Welcome → Toggle_Labels (Changed the warehouse in inspector to “Canvas_Labels” and make sure it’s “GameObject.SetActive”
3.	Factory Floor Lighting	Floor lightmap detail was visibly flatter than adjacent surfaces.	Warehouse → Floor, (Inspector) in the Lightmapping section, change the “scale in Lightmap” to ‘1’ from ‘0.1’.
4.	Box Pile Shadows	Back-left box pile lacked baked shadows consistent with other piles.	The “Pile 3” boxes in the hierarchy, just tick the box for static in the inspector and baked it.

5.	Console Lighting	Console appeared unlit by the baked light directly above it.	Surrounded the console with light probes.
6.	Magnet Visual Feedback	No indicator showed whether the magnet was active or lifting a box.	• Magnet active indicator: ControlsSetup → Controls → Center → Controls → Lever, drag “Magnet” into “On Lever Activate()” and choose “MeshRenderer” → “Material material”, the material can be chosen after putting the function. • Lifting a box indicator: MagnetSetup → Magnet → Socket, add particle effect in “Select Entered (SelectEnterEventArgs)” and set “MagnetParticleSystem.PlayMagnetParticles” function.
7.	Instruction Lock	Ray or direct interactors remained usable during onboarding, allowing console interaction before instructions were read.	Canvas_Main → Welcome → Text (TMP), add 3 more “On Complete ()” which were • LeftRay = GameObject.SetActive (uncheck) • RightRay = GameObject.SetActive (uncheck) • ControlsSetup = GameObject.SetActive (check)

1.3 Part B – Additional Feature: Safety Equipment Check Station

The group selected the “Safety Equipment Check Station” as its custom feature. At a dedicated station within the factory, the trainee can pick up and correctly wear a hard hat which is the

helmet in Unity. A checklist UI panel tracks which items have been correctly equipped in real time.

Rationale

This feature was chosen for three reasons. First, PPE compliance is directly relevant to factory training and reinforces the pedagogical purpose of the simulation rather than being a novelty add-on. Second, it exercises core XR interaction concepts required by the assignment such as grab interactions, socket-based placement, contextual UI feedback, and particle-based confirmation while remaining achievable to a high standard of polish within the project timeline. Third, its Low-Medium suggested complexity allows the group to focus effort on execution quality rather than spreading effort thin across a more complex system, in line with the brief's note that a simple feature executed excellently can score as well as a more ambitious one.

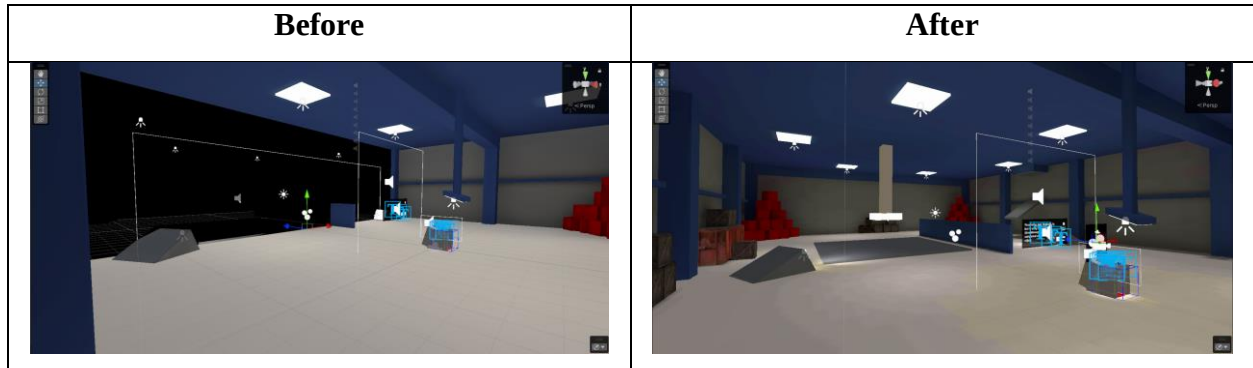
Feature Design

Interaction flow: the trainee approaches the station and sees PPE items resting on racks. Grabbing an item and bringing it near its socket triggers a highlight outline on the socket which releasing near the socket snaps the item into place with a satisfying particle burst if the correct item is placed in the socket. The checklist UI updates each row from white text to green colour text with strikethrough. A “Reset” button on the station returns all settings to default such as the hands' rays, snap turning and vignette effect.

2.0 Repair Documentation (Part A)

The following before and after evidence documents each of the seven Part A repairs. Screenshots should be captured from the same camera angle before and after each fix wherever possible, to make the contrast clear.

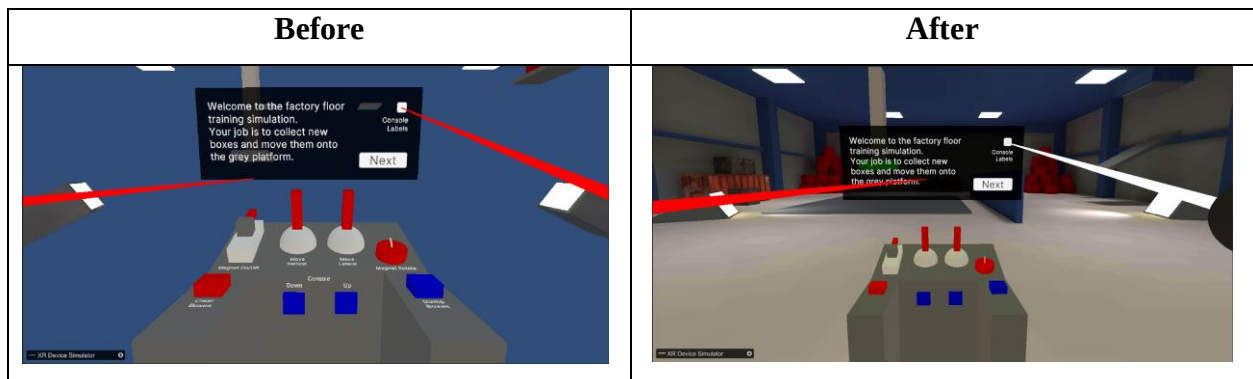
Item 1 – Onboarding Fade Screen



Issue: Fade screen positioned mid-scene, not covering the view.

Fix implemented: Fade Canvas reparented to camera, resized to fill field of view.

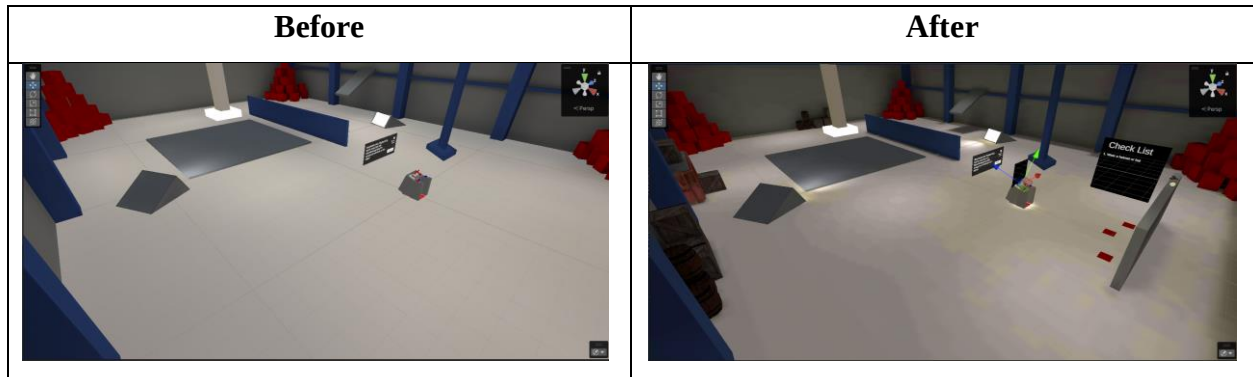
Item 2 – Console Labels Toggle



Issue: Toggle hides the building.

Fix implemented: Toggle now hides only label elements.

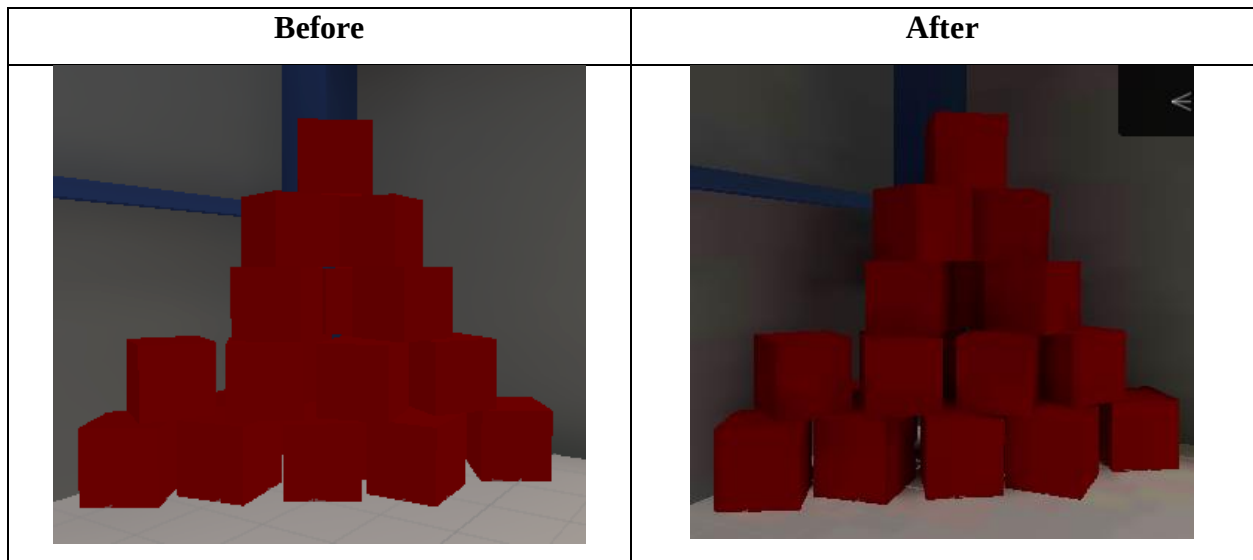
Item 3 – Factory Floor Lighting



Issue: Floor lightmap details are flatter than surroundings.

Fix implemented: Lightmap resolution increased and re-baked.

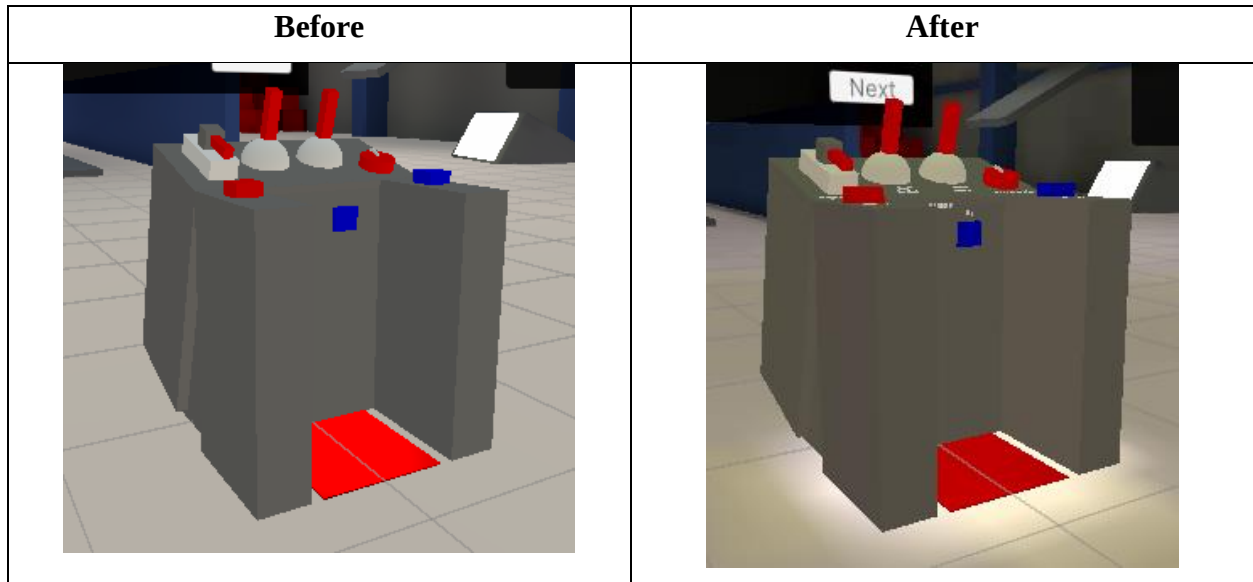
Item 4 – Box Pile Shadows



Issue: Back-left pile missing baked shadows.

Fix implemented: All piles marked Lightmap Static and re-baked.

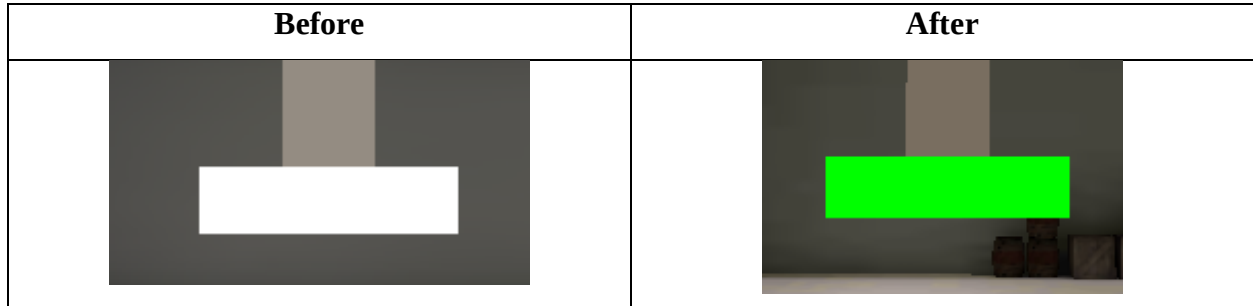
Item 5 – Console Lighting



Issue: Console dark under overhead baked light.

Fix implemented: Light probes added that console correctly lit while light stays baked.

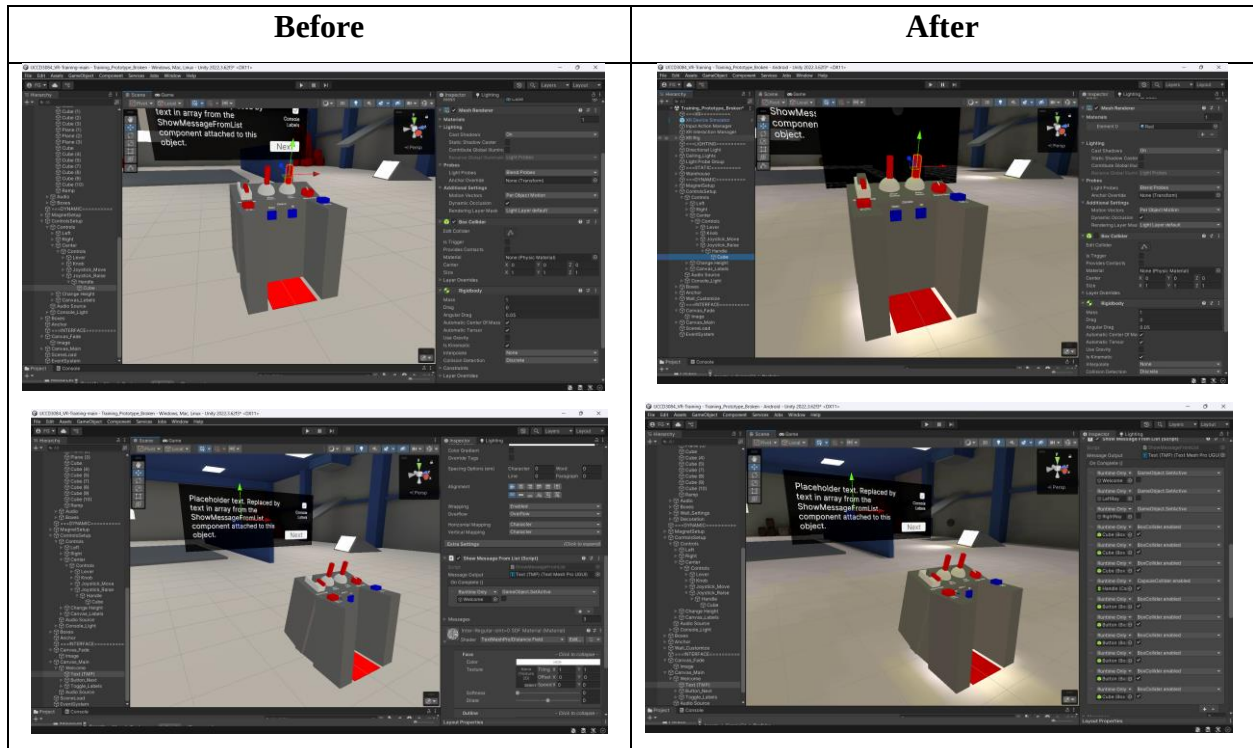
Item 6 – Magnet Visual Feedback



Issue: No indicator of magnet active or inactive state.

Fix implemented: Colour-swap and particle effect added for lifting states.

Item 7 – Instruction Lock



Issue: Console usable during onboarding instructions and rays still exist after instructions.

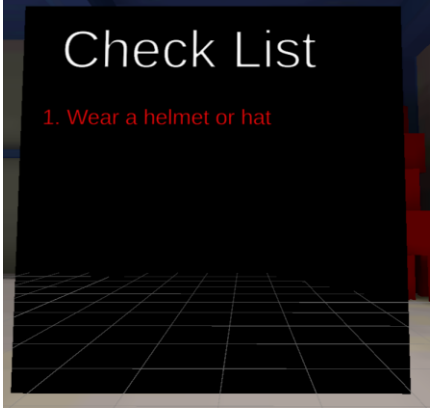
Fix implemented: Disabled each collider for the button and levers until instructions are dismissed. Disabled the rays after welcome message is dismissed.

3.0 Part B Documentation – Safety Equipment Check Station

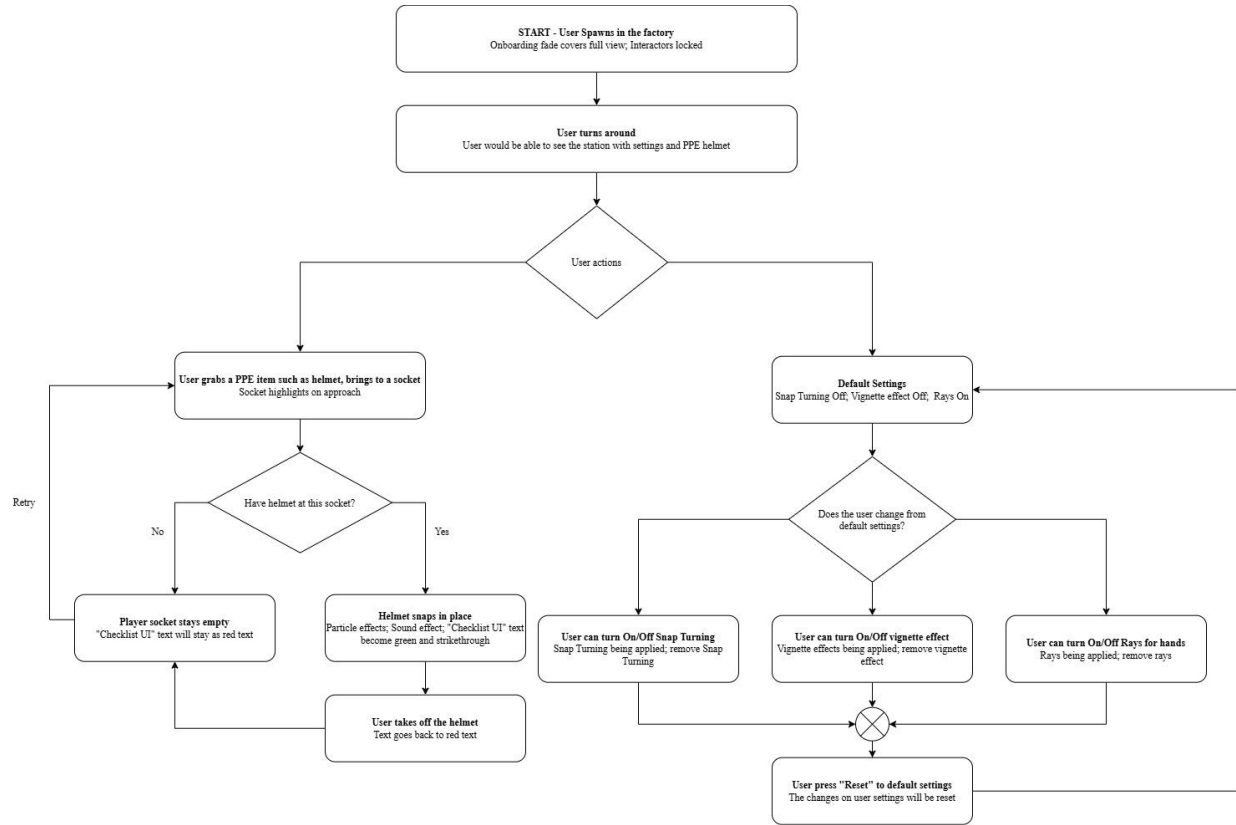


When the player starts the training, the player can turn around and see “Safety Equipment Check Station”. The hats and helmets will be hooked onto the wall when the player has spawn. For realism, we have added helmet as our PPE equipment for “Part B”. The anchor in front of the hooks was to make the player stand in front of the hats and helmets to be pick up easier and faster navigation. Next, the anchor in front of the mirror was to make the player be able to stand in front of it to see the player’s head is wearing a hat or helmet. On the right side of the wall, there are settings that the player can interact with their hands or rays. The snap turning and vignette is to help the player to reduce their motion sickness. The “Left and Right Ray” was to help the player to be able to use their rays again after they finish clicking through the instructions since it will disable it. The buttons can interact with their rays and hands. The player can also use their rays on the factory floor to navigate around the factory.

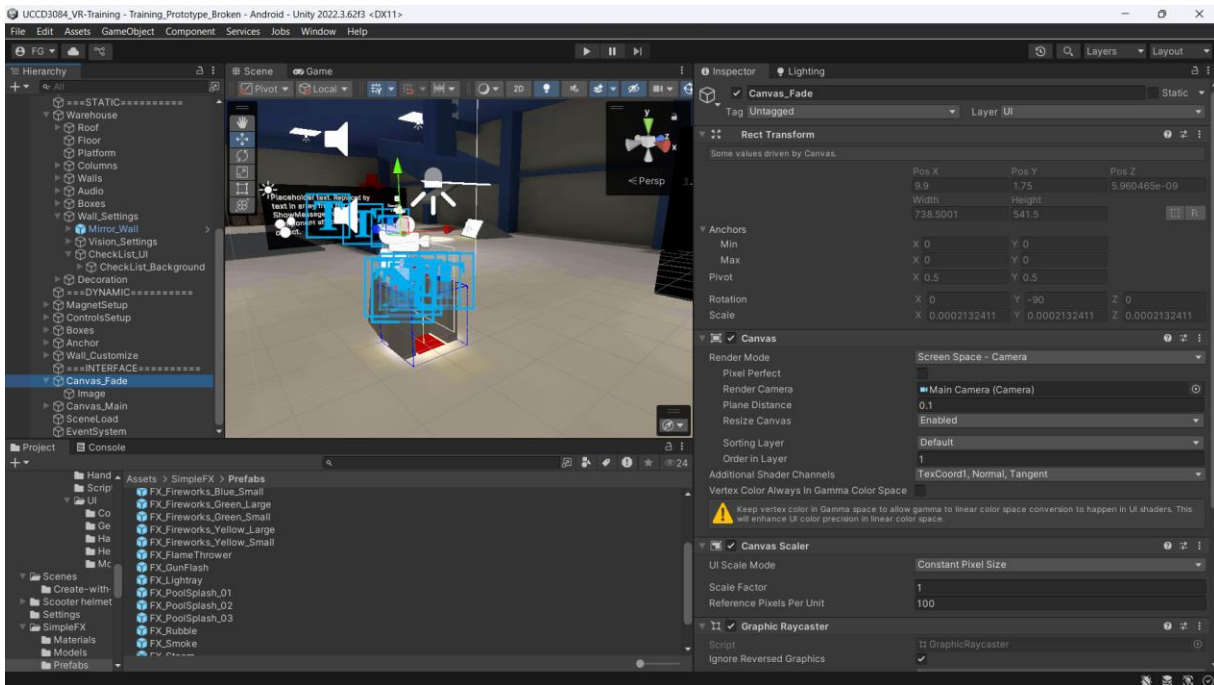
3.1 UI Elements

When the player has not wear a helmet or hat, the text will stay as red text.	When the player has worn a hat, the text will become green and strikethrough with sound effect indicate that the player had successfully worn a helmet or hat.
 A screenshot of a 'Check List' UI element. The title 'Check List' is in white. Below it, the first item '1. Wear a helmet or hat' is written in red text. The background is a dark grid pattern.	 A screenshot of a 'Check List' UI element from a first-person perspective. The title 'Check List' is in white. Below it, the first item '1. Wear a helmet or hat' is written in green text with a strikethrough. The background shows a dark, blocky environment. A small 'XR Device Simulator' logo is visible in the bottom left corner.

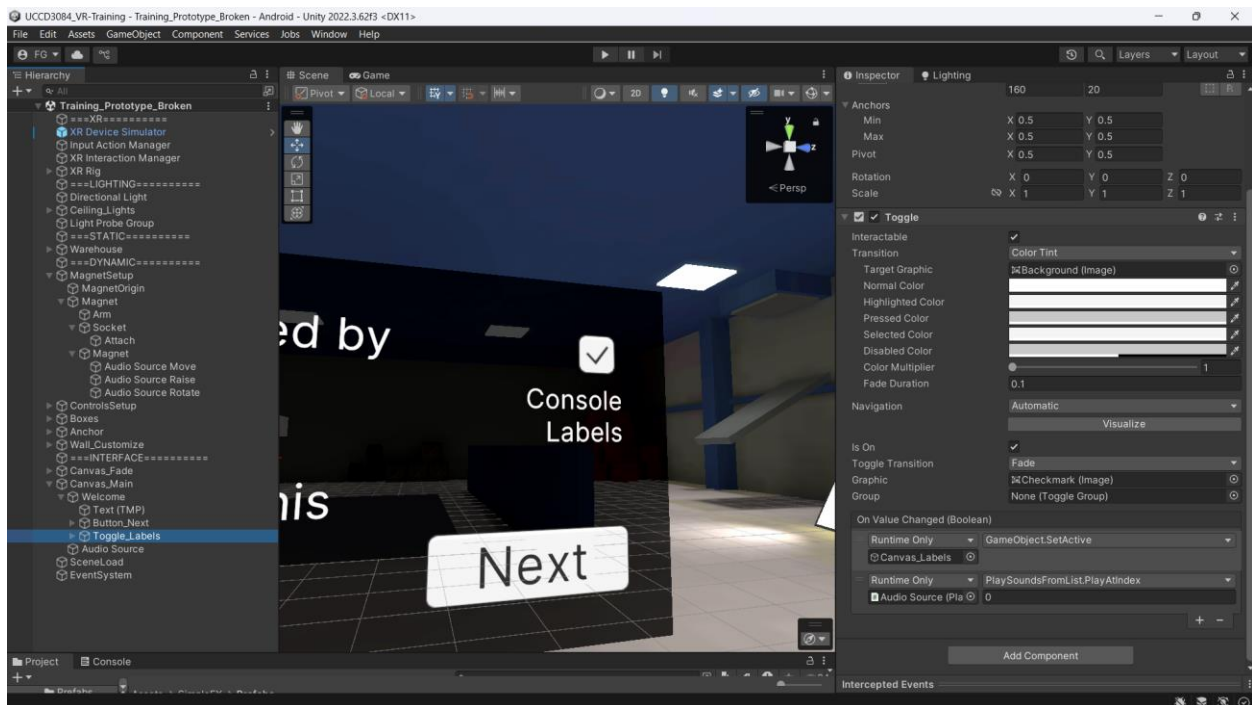
4.0 Visual Diagrams



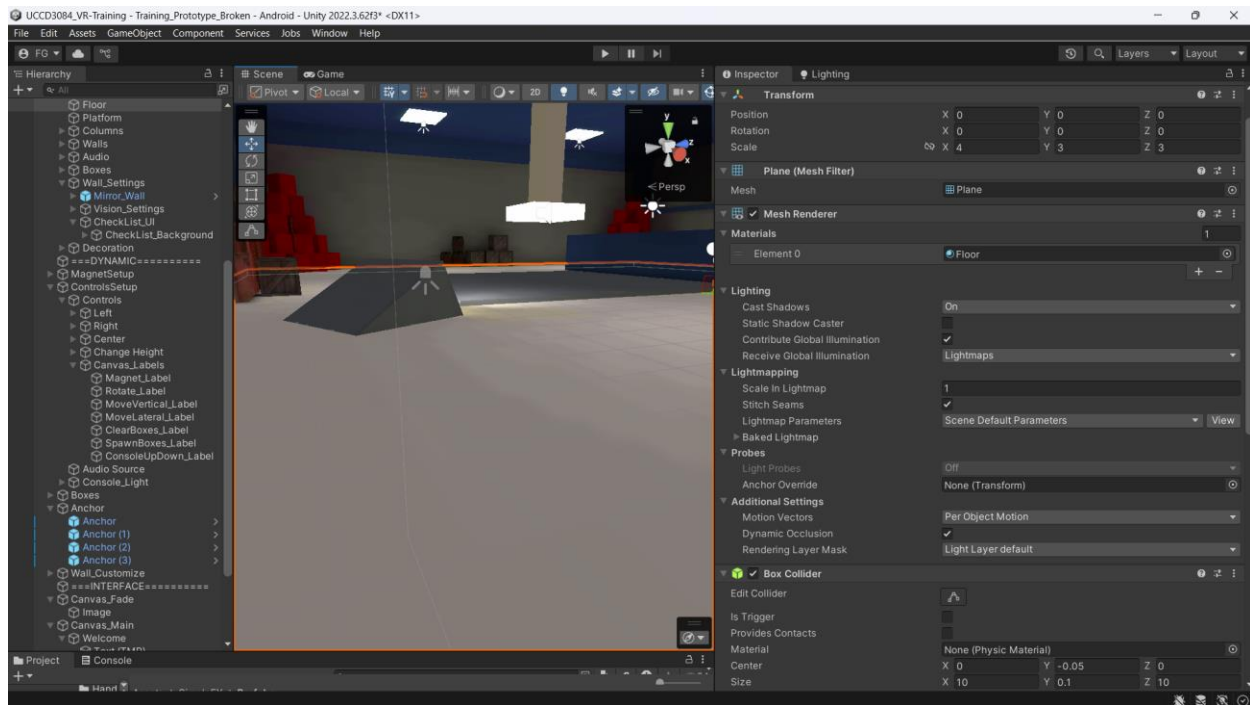
4.1 Part A - Repair logic or Scene Architecture



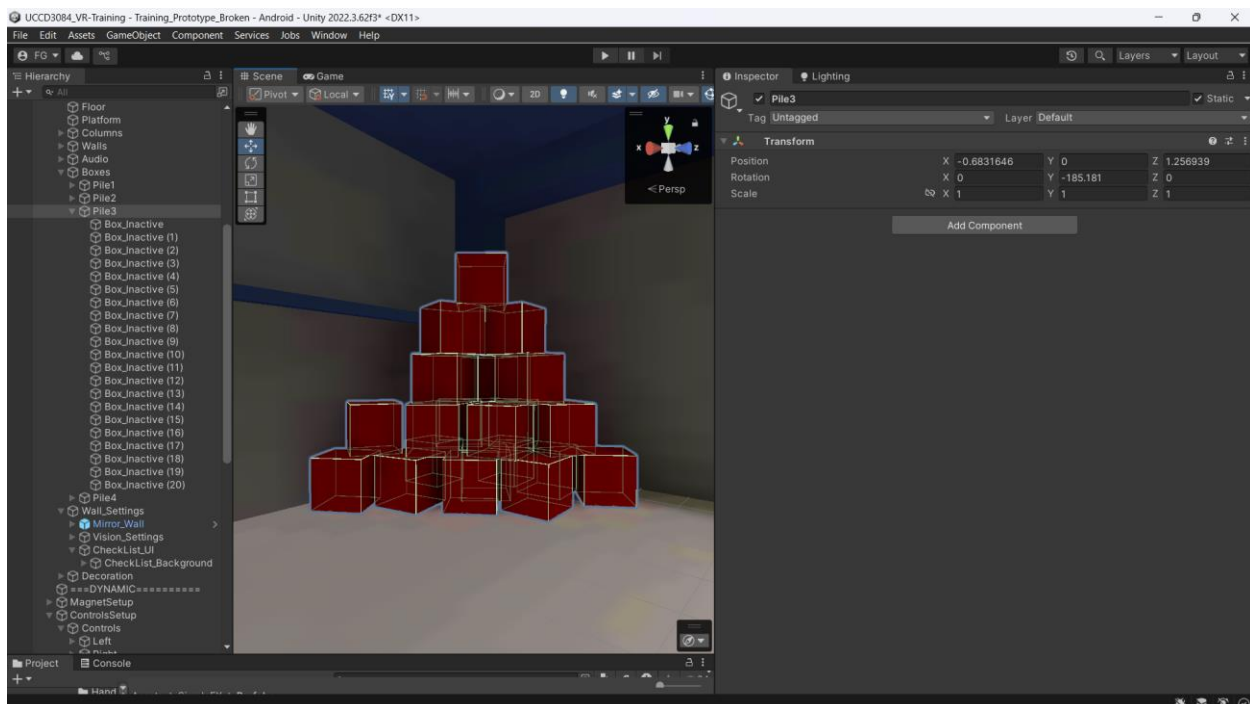
Caption: “Canvas_Fade component showing Render Mode and camera assignment, ensuring the fade screen covers the full field of view.”



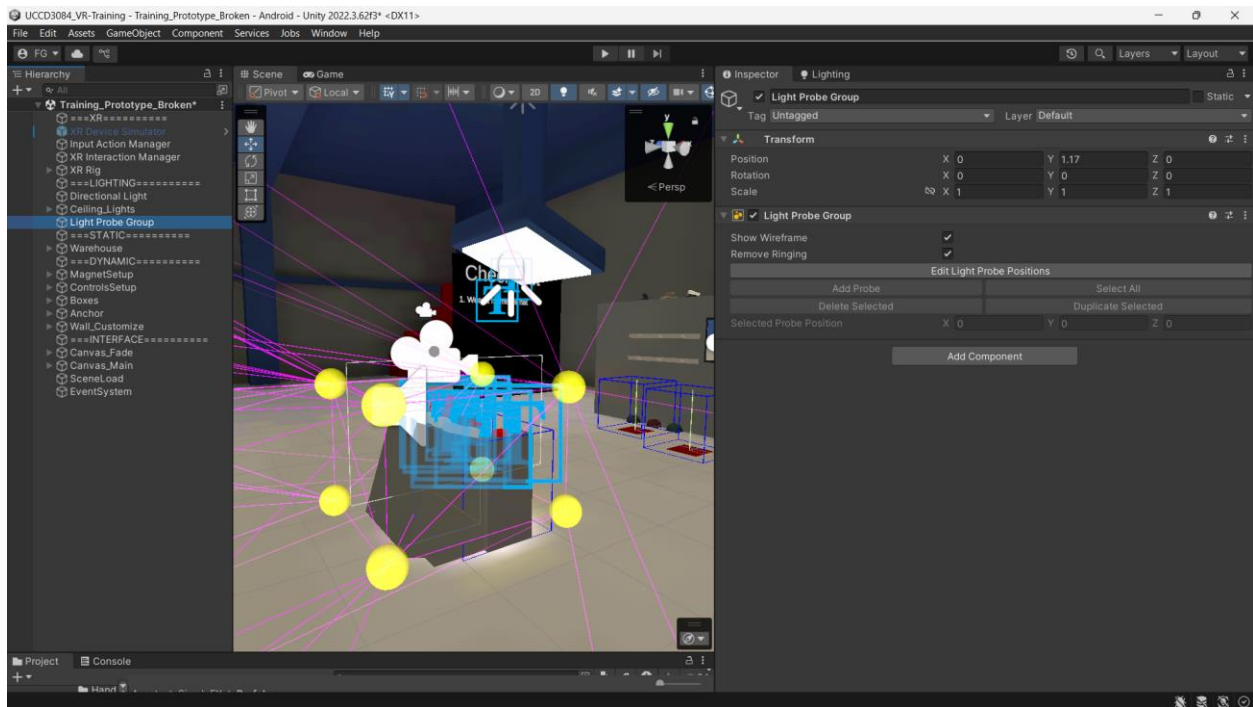
Caption: “Toggle_Labels On Value Changed() event repointed to the labels child object instead of the warehouse.”



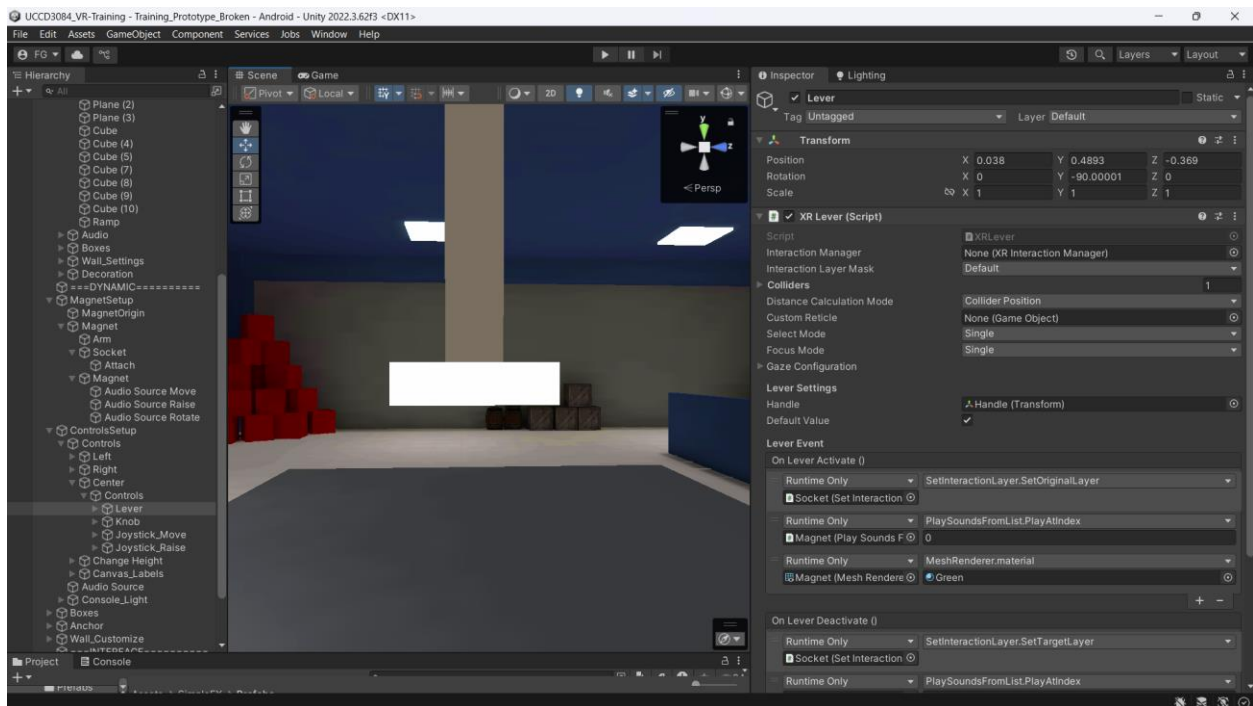
Caption: “Floor Mesh Renderer Lightmapping settings, showing increased Scale in Lightmap to match adjacent surfaces.”



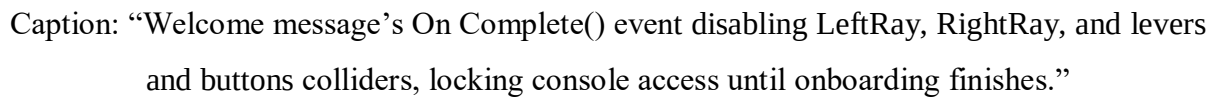
Caption: “Back-left box pile marked as Static and re-bake to ensure shadow consistency across all piles.”



Caption: “Light Probe Group positioned around the console so it samples correct baked illumination from the overhead light.”



Caption: “Magnet would turn green when the user switches the lever “On”, which when the magnet turns green it indicates the magnet could lift the box up.”

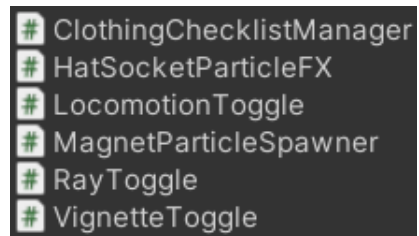


5. Team Contribution Summary

Name	Part A	Part B	Report Writing
Cheah Shao Feng	• UI – Onboarding Fade Screen • Toggle Logic Console Labels • Video Voice Recording	• Voice Recording	• Chapter 2
Kwek Xiao Qi	• Lighting – Factory Floor • Lighting – Box Piles	• SFX & VFX • Game Tester & Feedback	• Chapter 3
Yong Ler Xi	• Lighting – Console • Visual Feedback – Magnet State • Interaction – Instruction Lock	• Demo Recording • Code Debugging	• Chapter 1
Lai Keen Seng	• Code Debugging	• Safety Equipment Check Station • Final Code Compile	• Chapter 4

After we were done with our own tasks. We check each other's works and test the implemented items for "Part A" and "Part B" to make sure that it is working as intended. As well as the report to make sure that everything had met all the requirements.

Assisted AI-generated code:



```
# ClothingChecklistManager
# HatSocketParticleFX
# LocomotionToggle
# MagnetParticleSpawner
# RayToggle
# VignetteToggle
```

ClothingChecklistManager.cs

Manages a list of PPE checklist items (by string ID), each linked to a TMP text label, When an XR interactor’s “Select Entered” or “Exited” event fires. For example, an item is put on or taken off, it marks the matching checklist entry with a strikethrough and green colour, or reverts it if removed. This is core logic for “Part B (Safety Equipment Check Station)” which is the checklist UI feedback.

HatSocketParticleFX.cs

Plays a particle effect when an item such as the hard hat which is the helmet is placed into its socket, similar to “MagnetParticleSpawner” but with a fallback destroy timer if no “ParticleSystem” component is found on the instantiated prefab. This supports Part B’s required particle effect on a key interaction moment such as correct PPE placement.

LocomotionToggle.cs

Same “ON/OFF” button pattern again, this time controlling an “ActionBasedSnapTurnProvider” component’s enabled state plus an optional target GameObject. This supports the “Comfort Settings” requirement which is the snap turning toggle.

MagnetParticleSpawner.cs

Plays a particle effect at the magnet whenever it picks up a box, by hooking into the Socket Interactor’s “Select Entered” event. Instantiates the particle prefab at a spawn point or the object’s own position if none is set, plays it, then auto-destroys it once its lifetime ends. This supports “Item 6 (Magnet Visual Feedback)” for the requirement which is the colour-swap would be handled separately.

RayToggle.cs

A reusable “ON/OFF” button script that shows green “ON” or red “OFF” state and enables or disables a set of assigned GameObjects (the ray interactors) accordingly. It used for toggling ray visibility or interaction. This is relevant to “Item 7 (Instruction Lock)” if wired to the ray objects, or general interactor control.

VignetteToggle.cs

Same “ON” or “OFF” button pattern as RayToggle, but generalized to switch either a component’s enabled flag or a GameObject’s active state. It used here for the comfort vignette

setting. This supports the “Comfort Settings” requirement for vignette toggle under “Overall VR Experience Requirements”.

Credits for Assets:

Asset Name: PBR Barrels and Crates

Username: Integrity Software & Games

Email: assets@integrity-sg.com

License agreement: Standard Unity Asset Store EULA

License type: Extension Asset

URL: <https://assetstore.unity.com/packages/3d/props/exterior/pbr-barrels-and-crates-34608>

Asset Name: Simple FX – Cartoon Particles

Username: Synty Studios

Email: support@syntystudios.com

License agreement: Standard Unity Asset Store EULA

License type: Extension Asset

URL: <https://assetstore.unity.com/packages/vfx/particles/simple-fx-cartoon-particles-67834>

Asset Name: Scooter helmet

Username: lowpoly89

Email: olegsolyanik89@gmail.com

License agreement: Standard Unity Asset Store EULA

License type: Extension Asset

URL: <https://assetstore.unity.com/packages/3d/props/clothing/accessories/scooter-helmet-285857>